ANSWER KEY & MARKING SCHEME SET 08

Full Syllabus (Chapters 1–11) | Class 7 Science | Maximum Marks: 80

SECTION — A (Objective Type) — Answers — 20 × 1 = 20

Q.	Correct Option	Brief Reasoning
1	**(a)** basic	they neutralise the excess acid
2	**(b)** a mild base such as an antacid	neutralises stomach acid
3	**(b)** a gap in the path	current cannot flow
4	**(d)** a long line and a short line	long = positive
5	**(b)** kerosene	reacts with air and water
6	**(a)** poor conductors of heat and electricity	except graphite
7	**(a)** chemical change	breakdown into simpler substances
8	**(d)** Combustion	burning
9	**(b)** a blood-related health problem	needed by the body
10	**(a)** Growth of facial hair	moustache and beard
11	**(b)** day	differences in specific heat
12	**(b)** recharge groundwater	conserves water
13	**(c)** 100 km	distance = speed × time
14	**(b)** sundial	Sun's shadow
15	**(b)** carbon dioxide	test for CO_2
16	**(c)** basic in nature	neutralises acid
17	**(b)** energy	uses oxygen
18	**(c)** make its own food	depends on others
19	**(d)** reflects sunlight	non-luminous
20	**(d)** Earth	natural satellite

SECTION — B (Very Short Answer) — Answers — 8 × 2 = 16

Q.	Expected Answer / Value Points	Marks
21	They are bases that neutralise excess stomach acid (2).	2
22	It melts and breaks the circuit when current is too high (2).	2
23	Oxygen and carbon (2).	2
24	Builds stamina and improves mood (2).	2
25	Hot air is lighter and moves upward by convection (2).	2
26	54 km/h = 15 m/s so 20 m/s is faster (2).	2
27	Absorbs water and salts and forms stool (2).	2
28	To release energy from food for growth (2).	2

SECTION — C (Short Answer) — Answers — 8 × 3 = 24

Q.	Expected Answer / Value Points	Marks
29	No; rubber is an insulator (3).	3
30	Cold light through chemical reactions (3).	3
31	Infiltration fills aquifers (3).	3
32	Separate pathways (1.5); rings keep it open (1.5).	1½+1½
33	Allows vision (1.5); needed for photosynthesis (1.5).	1½+1½
34	Months follow lunar cycle (1.5); festivals use tithis (1.5).	1½+1½
35	Crush fallen petals in warm water, filter (1.5); acids give pink, bases give green or dull colour (1.5).	1½+1½
	OR: Antacid for indigestion (1); lime in acidic soil (1); baking soda on ant bite (1).	1+1+1

36	Root hairs absorb (1); xylem carries upward (1); transpiration pulls (1).	1+1+1
	OR: Flat, thin blade with large area (1); chlorophyll in cells (1); stomata and veins (1).	1+1+1

SECTION — D (Long Answer) — Answers

4 × 5 = 20

Q.	Expected Answer / Value Points	Marks
37	(i) Nail, wire, lead (1). (ii) Straw (1). (iii) Graphite is a conducting non-metal (1). (iv) Electrical wiring (2).	1+1+1+2
	OR: (i) Mixture (1). (ii) Steel, brass, bronze (2). (iii) Stronger, resist corrosion (2).	1+2+2
38	(i) Needs nutrients for growth (1). (ii) Iron, protein (1). (iii) Eat balanced meals (1). (iv) Doctor/dietitian (2).	1+1+1+2
	OR: (i) Desire to belong (1). (ii) Positive habits or harmful ones (2). (iii) Think, say no, seek guidance (2).	1+2+2
39	(i) Bob on a thread (1). (ii) Length (2). (iii) Clocks (2).	1+2+2
	OR: (i) Change of position (1). (ii) Linear, circular, periodic (2). (iii) Speedometer (2).	1+2+2
40	(i) Travels in vacuum (2). (ii) Through space (2). (iii) Sound needs a medium (1).	2+2+1
	OR: (i) Dark patch vs picture (2). (ii) Erect virtual vs inverted real (2). (iii) Pinhole image (1).	2+2+1

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

MASTERMIND